

ENVS 193DS Final

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Problem 1:

a) Transparent Statistical Methods

In part 1 they used a logical regression. They are comparing a binary variable and a continuous variable. In part 2 they are using a chi-square test, measuring the association between two variables.

b) Suggestions for Rewriting

Part 1:

American Avocets are more likely to use micro habitats at certain distances from the edge of the water. Thus proving that the distance to the water edge influences habitat use patterns.

Part 2:

Different bird species differ in their habitat. American Avocets, Foster's Terns, and Black necked Stilts all show different habitat preferences across wetlands.

c) Further Analysis

An additional variable that may be influencing the American Avocet micro habitat use is the water depth. This is a continuous variable that is measuring depth by centimeters using a meter stick. Depth of water may affect the availability of food for the birds. Eating a variety of aquatic animals, the depth of water will heavily affect their ability to catch food and the presence of food in general.

Problem 2:

a)

```
library(tidyverse)
library(janitor)

# read in data
nests <- read_csv("nest_data_final.csv")
```

```

# clean and wrangle data
nests_clean <- nests %>%
  clean_names() %>%
  select(ant_species, height_cm, alive) %>%
  drop_na() %>%
  mutate(
    alive = factor(alive, levels = c("D", "A")),
    ant_species = factor(
      ant_species,
      levels = c("RRB", "AB", "BBR")
    )
  ) %>%

  select(ant_species, height_cm, alive) %>%

  filter(ant_species %in% c("AB", "RRB", "BBR")) %>%

  # remove missing values
  drop_na(ant_species, height_cm, alive) %>%

  mutate(

    # correctly convert alive to numeric
    alive = ifelse(alive == 1 | alive == "1", 1, 0),

    # make ant species a factor
    ant_species = factor(
      ant_species,
      levels = c("RRB", "AB", "BBR")
    ),

    # full species names
    ant_species_full = case_when(
      ant_species == "RRB" ~ "Crematogaster sjostedti",
      ant_species == "AB" ~ "Crematogaster mimosae",
      ant_species == "BBR" ~ "Crematogaster nigriceps"
    )
  )

# check categories
table(nests_clean$ant_species)

```

```
RRB AB BBR
209 16 45
```

```
table(nests_clean$alive)
```

```
0
270
```

```
# Model 1
model_1 <- glm(
  alive ~ height_cm + ant_species,
  data = nests_clean,
  family = binomial
)

# Model 2
model_2 <- glm(
  alive ~ height_cm * ant_species,
  data = nests_clean,
  family = binomial
)
```

b) 1 indicates that a tree contained a nest, 0 means the tree is “available” meaning there is no nest

c) Tree height is a continuous variable measured in centimeters or meters. The probability of ant nests becomes greater as height increases. Greater heights provide greater protection from predators. This information can be found in line 95-99.

d)

```
# create table of models

models_table <- tibble(
  model_number = c("Model 1", "Model 2"),

  predictors_included = c(
    "Tree height + ant species",
    "Tree height + ant species + tree height × ant species"
  ),
```

```

interaction_term_present = c(
  "No",
  "Yes"
),

model_description = c(
  "Tests whether tree height and ant
  species independently affect the
  probability of nest occurrence.",

  "Tests whether the effect of tree
  height on the probability of nest
  occurrence differs among ant species."
)
)

# display table
models_table

```

```

# A tibble: 2 x 4
  model_number predictors_included interaction_term_pre~1 model_description
  <chr>         <chr>              <chr>              <chr>
1 Model 1      Tree height + ant speci~ No          "Tests whether t~
2 Model 2      Tree height + ant speci~ Yes         "Tests whether t~
# i abbreviated name: 1: interaction_term_present

```

e)

```

#read in data
nests_clean <- nests_clean %>%
  mutate(
    alive = if_else(alive == "yes", 1, 0)
  )
# Model 1
model_1 <- glm(
  alive ~ height_cm + ant_species,
  data = nests_clean,
  family = binomial,
  control = glm.control(maxit = 100)
)

```

```
# Model 2
model_2 <- glm(
  alive ~ height_cm * ant_species,
  data = nests_clean,
  family = binomial,
  control = glm.control(maxit = 100)
)
```

f)

```
AIC(model_1,model_2)
```

```
      df AIC
model_1  4   8
model_2  6  12
```

The best model as determined by Akaike's Information Criterion (AIC) was model 1, the interaction between tree height and ant species.

g)

```
library(DHARMA)

# read data
nests <- read_csv("nest_data_final.csv")

# clean data
nests_clean <- nests %>%

  clean_names() %>%

  select(ant_species, height_cm, alive) %>%

  drop_na(ant_species, height_cm, alive) %>%

  mutate(

    # convert A/D into 1/0
    alive = if_else(
      trimws(toupper(alive)) == "A",
      1,
```

```

      0
    ),

    ant_species = factor(
      ant_species,
      levels = c("RRB", "AB", "BBR")
    )
  )

# IMPORTANT: check response variable
table(nests_clean$alive)

```

```

0    1
16 283

```

```

# fit model
model_1 <- glm(
  alive ~ height_cm + ant_species,
  data = nests_clean,
  family = binomial
)

# check model
summary(model_1)

```

Call:

```
glm(formula = alive ~ height_cm + ant_species, family = binomial,
    data = nests_clean)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	6.155e+00	2.346e+00	2.624	0.00869 **
height_cm	-3.374e-03	7.982e-03	-0.423	0.67249
ant_speciesAB	1.625e+01	7.235e+03	0.002	0.99821
ant_speciesBBR	1.599e+01	4.342e+03	0.004	0.99706

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

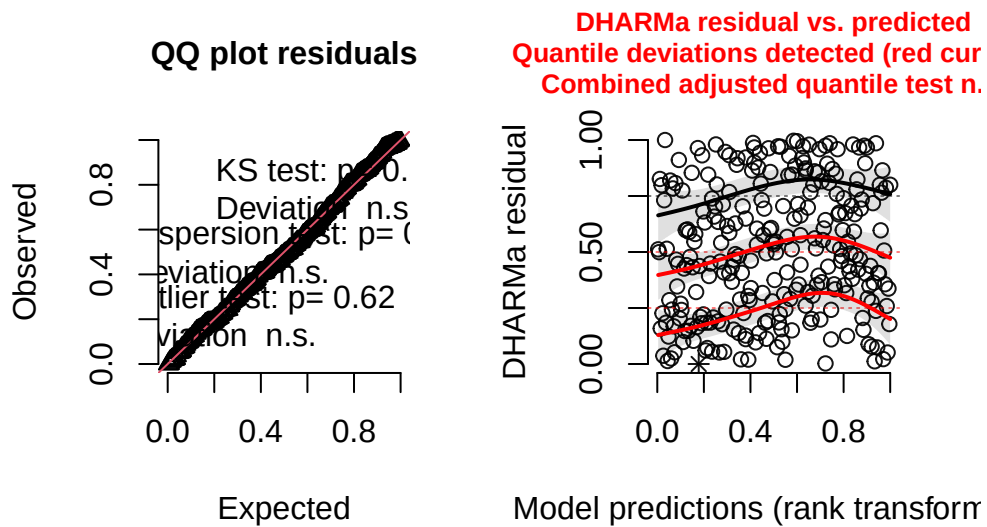
Null deviance: 13.193 on 269 degrees of freedom
 Residual deviance: 12.521 on 266 degrees of freedom
 (29 observations deleted due to missingness)
 AIC: 20.521

Number of Fisher Scoring iterations: 20

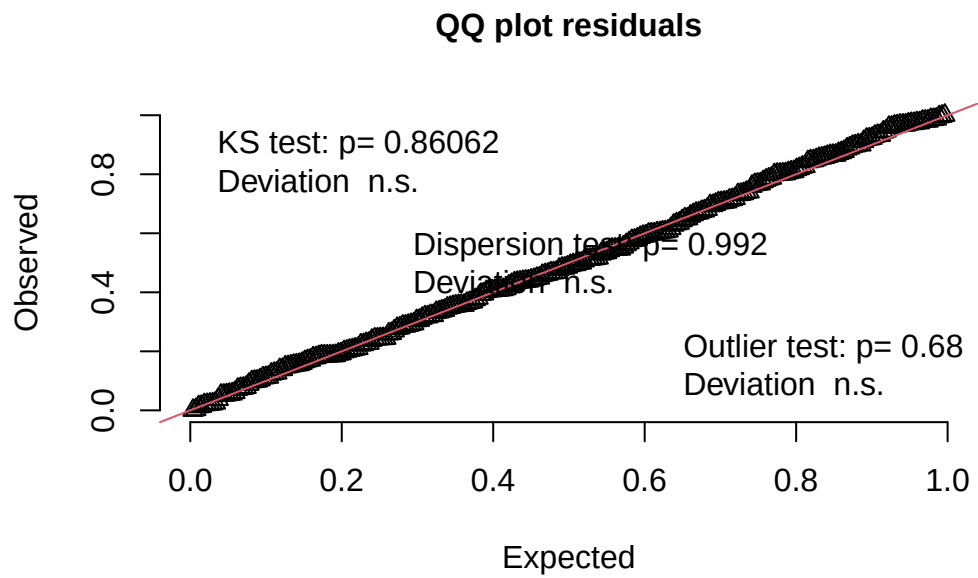
```
# simulate residuals
model_1_res <- simulateResiduals(
  fittedModel = model_1,
  plot = FALSE
)

# diagnostic plots
plot(model_1_res)
```

DHARMa residual



```
# tests
testUniformity(model_1_res)
```

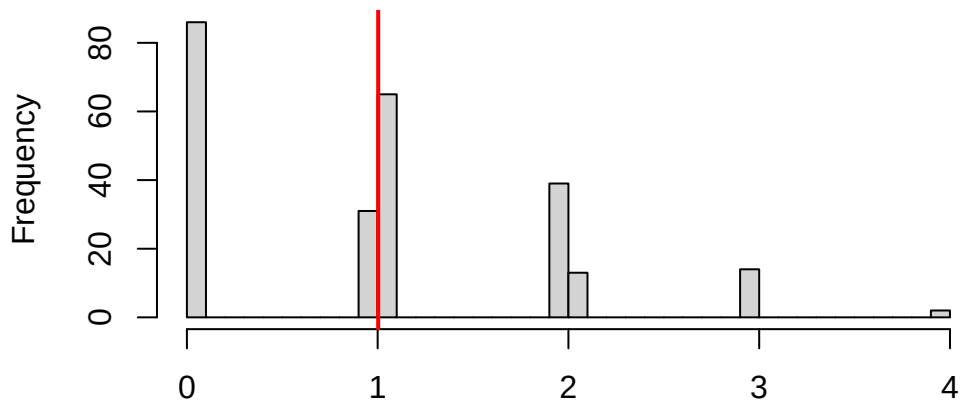


Asymptotic one-sample Kolmogorov-Smirnov test

```
data: simulationOutput$scaledResiduals  
D = 0.036682, p-value = 0.8606  
alternative hypothesis: two-sided
```

```
testDispersion(model_1_res)
```


**DHARMa nonparametric dispersion test via sd of
residuals fitted vs. simulated**



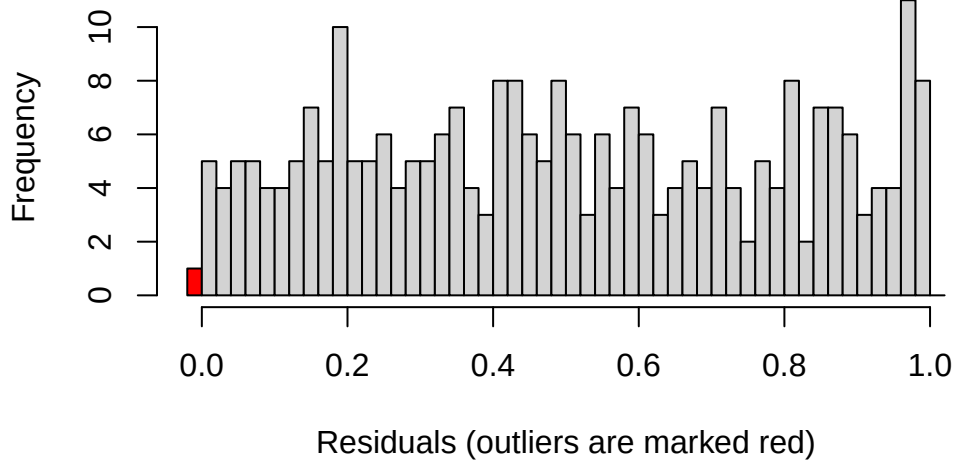
Simulated values, red line = fitted model. p-value (two.sided) = 0.992

DHARMa nonparametric dispersion test via sd of residuals fitted vs.
simulated

```
data: simulationOutput
dispersion = 1.0046, p-value = 0.992
alternative hypothesis: two.sided
```

```
testOutliers(model_1_res)
```

Outlier test n.s.



DHARMa bootstrapped outlier test

```
data: model_1_res
outliers at both margin(s) = 1, observations = 270, p-value = 0.64
alternative hypothesis: two.sided
percent confidence interval:
 0.000000000 0.007407407
sample estimates:
outlier frequency (expected: 0.00144444444444444 )
                                0.003703704
```

h)

```
# read in data
nests <- read_csv("nest_data_final.csv")

# clean and wrangle data
nests_clean <- nests %>%

  clean_names() %>%
```

```

select(ant_species, height_cm, alive) %>%

drop_na(ant_species, height_cm, alive) %>%

mutate(

  # convert alive/dead to binary values
  alive = if_else(
    trimws(toupper(alive)) == "A",
    1,
    0
  ),

  # make species a factor
  ant_species = factor(
    ant_species,
    levels = c("RRB", "AB", "BBR")
  ),

  # full species names for plotting
  ant_species_full = case_when(
    ant_species == "RRB" ~ "Crematogaster sjostedti",
    ant_species == "AB" ~ "Crematogaster mimosae",
    ant_species == "BBR" ~ "Crematogaster nigriceps"
  )
)

# fit selected model
model_1 <- glm(
  alive ~ height_cm + ant_species,
  data = nests_clean,
  family = binomial
)

# create sequence of heights for smooth predictions
height_seq <- seq(
  min(nests_clean$height_cm),
  max(nests_clean$height_cm),
  length.out = 100
)

# create prediction dataset

```

```

new_data <- expand.grid(
  height_cm = height_seq,
  ant_species = levels(nests_clean$ant_species)
)

# add full species names
new_data <- new_data %>%
  mutate(
    ant_species_full = case_when(
      ant_species == "RRB" ~ "Crematogaster sjostedti",
      ant_species == "AB" ~ "Crematogaster mimosae",
      ant_species == "BBR" ~ "Crematogaster nigriceps"
    )
  )

# generate predictions with standard errors
preds <- predict(
  model_1,
  newdata = new_data,
  type = "link",
  se.fit = TRUE
)

# calculate predicted probabilities and 95% confidence intervals
new_data <- new_data %>%
  mutate(
    fit_link = preds$fit,
    se = preds$se.fit,

    lower_link = fit_link - 1.96 * se,
    upper_link = fit_link + 1.96 * se,

    fit = plogis(fit_link),
    lower = plogis(lower_link),
    upper = plogis(upper_link)
  )

# create plot
ggplot() +

  # underlying data
  geom_jitter(

```

```

data = nests_clean,
aes(
  x = height_cm,
  y = alive,
  color = ant_species
),
width = 5,
height = 0.03,
alpha = 0.5,
size = 2
) +

# confidence interval ribbon
geom_ribbon(
  data = new_data,
  aes(
    x = height_cm,
    ymin = lower,
    ymax = upper
  ),
  fill = "skyblue",
  alpha = 0.3
) +

# prediction line
geom_line(
  data = new_data,
  aes(
    x = height_cm,
    y = fit
  ),
  color = "darkblue",
  linewidth = 1
) +

# separate panels for each species
facet_wrap(~ant_species_full) +

# axis labels
labs(
  x = "Tree height (cm)",
  y = "Predicted probability of nest occupancy"
)

```

```

) +

# custom colors
scale_color_manual(
  values = c(
    "#1b9e77",
    "#d95f02",
    "#7570b3"
  )
) +

# clean theme
theme_classic() +

# remove legend
theme(
  legend.position = "none"
)

```

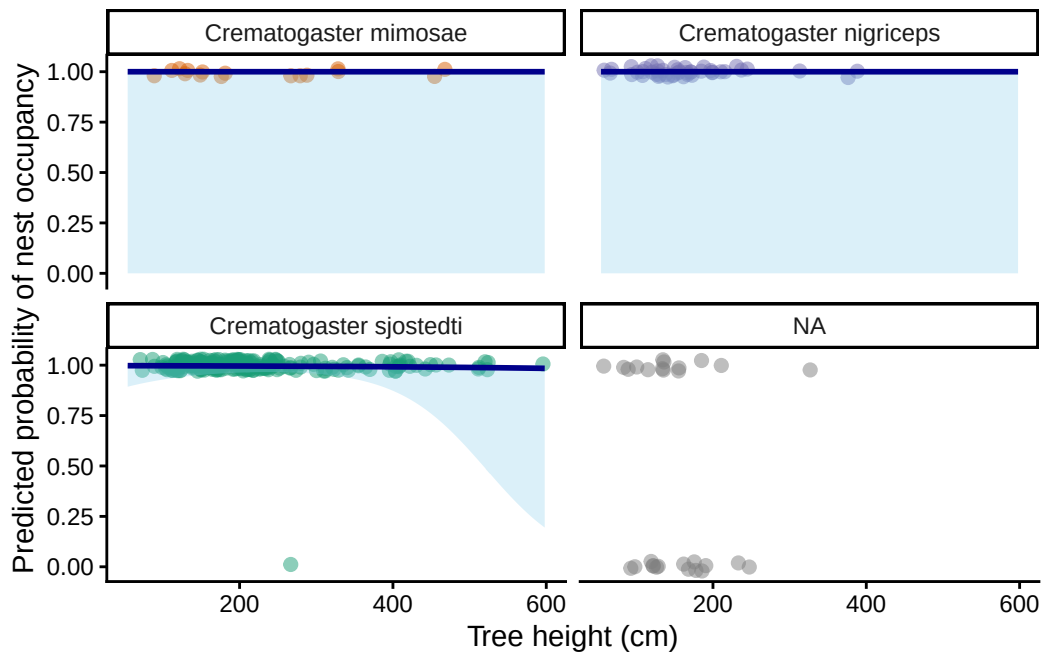


Figure 1: Probability of Survival Dependent on Nest Height of three different Ant Species. Logistic regression curves show the probability that a nest was alive across different tree heights. The orange points represents the *Crematogaster mimosae*, the majority of this species nests at 200cm with a 1.00 probability rate. The purple represents *crematogaster nigriceps*,

they nest at around 200cm with a 1.00 probability. The green is *crematogaster sjostedti*, they have a wider range of nest height. Lujan, E., Nielsen, R., Short, Z., Wicks, S., Nderitu Watetu, W., Malingati Khasoha, L., Palmer, T. M., Goheen, J. R., & Alston, J. M. (2023)

j)

```
# create new data for prediction
new_data <- data.frame(
  height_cm = 600,
  ant_species = factor(
    c("RRB", "AB", "BBR"),
    levels = levels(nests_clean$ant_species)
  )
)

# calculate predicted probabilities
predictions <- predict(
  model_1,
  newdata = new_data,
  type = "response"
)

# combine predictions with species names
prediction_output <- cbind(new_data, predicted_probability = predictions)

# display output
prediction_output
```

	height_cm	ant_species	predicted_probability
1	600	RRB	0.9841796
2	600	AB	1.0000000
3	600	BBR	1.0000000

k) Figure one showed that all three species of ants mostly nest around 200cm and have a probability of nest occupancy of about 1.00. Both BBR and AB have a predicted probability of 1.000 at 600cm, this is because they are more aggressive species. RRB has a probability of 0.9841796. They have a lower probability because they are less aggressive. The logistic regression model indicated that both tree height and ant species influence the nest occurrence. Trees occupied by less aggressive ants tend to have more nests. This occurs because some acacia ants aggressively defend their habitat by attacking birds.

Problem 3

a) The visualizations in homework 3 are more clear and concise compared to homework 2. In homework 2, there were two different graphs that interpreted the results. One graph is jittered scatter plot that shows the spread of points. The other plot is a boxplot that shows the minimum and maximum values along with the median values. In homework 3, the plot is a combination of both plots. It shows the upper and lower quartiles, median values, and the spread of values. It gives a better sense into the distribution of points. Both homeworks are communicating the same data, but homework 3 does a better job of doing it more concisely.

b) Another analysis that could be run on this data is a one-way ANOVA to test how urchin gonad weight differs between experimental treatments. The response variable would be the gonad weight, which is a continuous quantitative variable. The predictor variable is the experimental treatment, which is a categorical variable.

This analysis would answer the question how different treatments affect urchin gonad weight. The ANOVA will compare the means of a continuous response to a categorical predictor.

c)

```
# Load package
library(car)

# Import dataset
urchins <- read_csv("urchin-experiment-data.csv")

# Clean dataset
urchins_clean <- urchins |>

  clean_names() |>

  select(experimental_treatment, weight_g) |>

  drop_na(weight_g) |>

  mutate(
    experimental_treatment = dplyr::recode(
      experimental_treatment,
      "c" = "Control",
      "n" = "Nutrient-enriched"
    )
  )

# Create QQ plots to assess normality
ggplot(urchins_clean,
  aes(sample = weight_g)) +
```



```

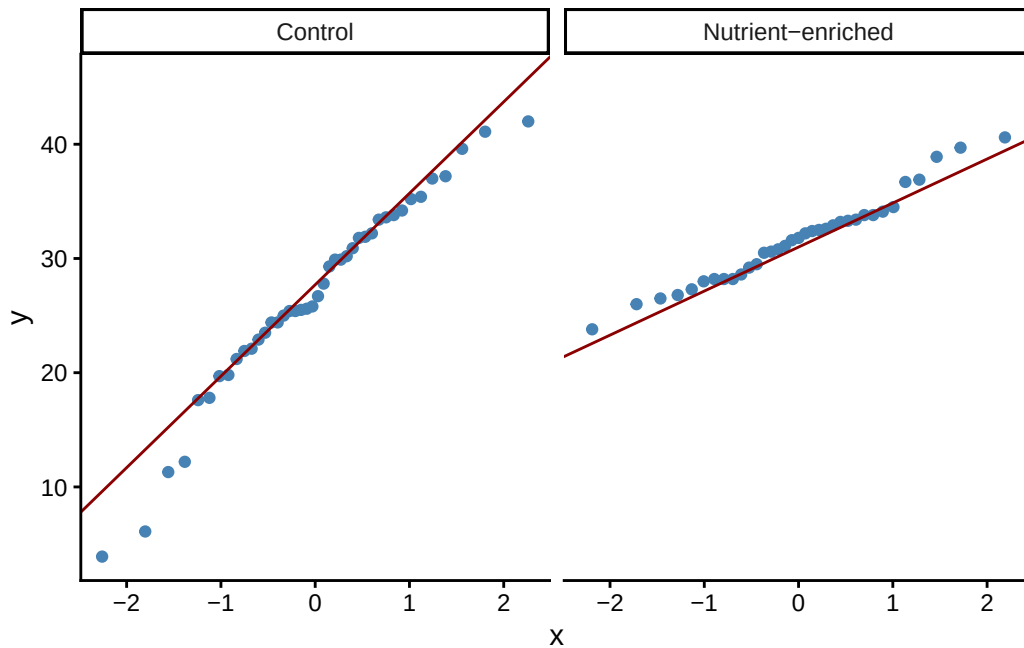
stat_qq(color = "steelblue") +

stat_qq_line(color = "darkred") +

facet_wrap(~ experimental_treatment) +

theme_classic()

```



```

# Test homogeneity of variance
leveneTest(
  weight_g ~ experimental_treatment,
  data = urchins_clean
)

```

Levene's Test for Homogeneity of Variance (center = median)

	Df	F value	Pr(>F)
group 1	1	13.664	0.0004136 ***
	75		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
# Run one-way ANOVA
urchin_anova <- aov(
  weight_g ~ experimental_treatment,
  data = urchins_clean
)
```

```
# Show ANOVA table
summary(urchin_anova)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
experimental_treatment	1	456	455.9	9.488	0.00289 **
Residuals	75	3603	48.0		

 Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

d)

```
# Create visualization for ANOVA results
ggplot(urchins_clean,
  aes(x = experimental_treatment,
      y = weight_g,
      fill = experimental_treatment)) +

# Add boxplots to show group distributions
geom_boxplot(
  width = 0.55,
  alpha = 0.75,
  outlier.shape = NA
) +

# Add underlying raw data
geom_jitter(
  width = 0.12,
  alpha = 0.7,
  size = 2,
  color = "black"
) +

# Add group means
stat_summary(
  fun = mean,
```

```

    geom = "point",
    size = 4,
    color = "red"
  ) +

  # Customize colors
  scale_fill_manual(
    values = c("steelblue4", "darkorange2")
  ) +

  # Label axes
  labs(
    title = "Effects of Experimental Treatment on Urchin Gonad Weight",
    x = "Experimental Treatment",
    y = "Urchin Gonad Weight (g)"
  ) +

  # Use non-default theme
  theme_classic() +

  # Remove unnecessary legend
  theme(
    legend.position = "none",
    plot.title = element_text(face = "bold", size = 14),
    axis.title = element_text(size = 12)
  )

```

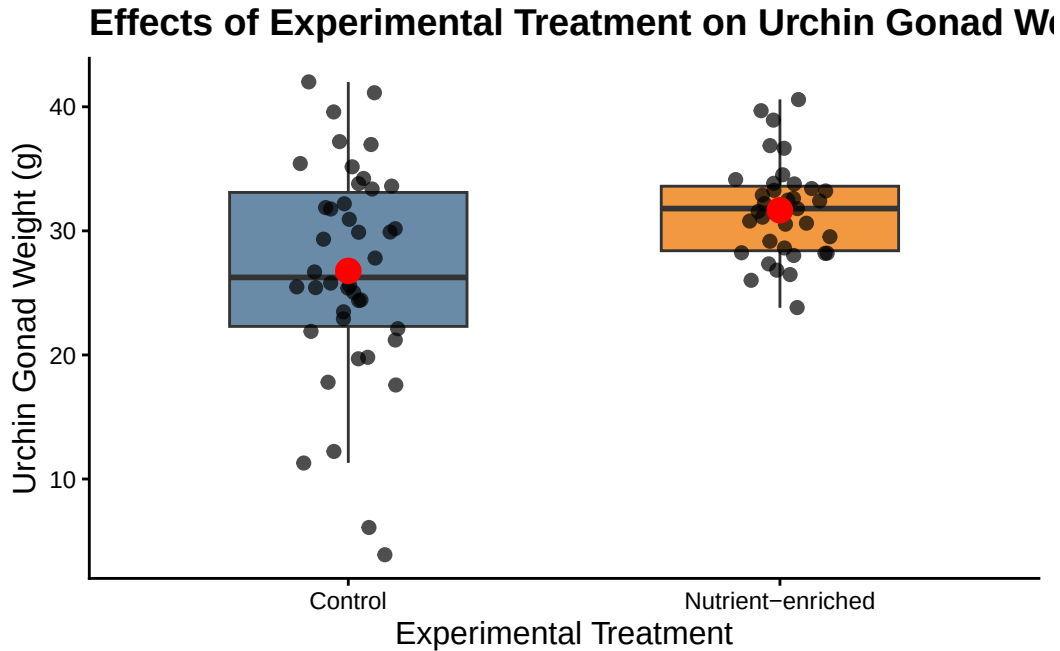


Figure 2: This plot demonstrates the effect of experimental treatment on sea urchin gonad weight in grams. The blue box represents the interquartile range for the control group. The median for this group is about 25g. The orange represents the nutrient enriched urchins. The median for this group being about 32g. The black jitter points represent the individual observations and the red points represents the mean gonad weight for each group.

f) The one-way ANOVA showed that the experimental treatment had a significant effect on urchin gonad weight (F , = 9.49, p = 0.0029). Urchins that were given the treatment had larger gonads than those in the control treatment. Levene's test showed unequal variances between groups (F , = 13.66, p < 0.001), meaning the homogeneity of variance assumption for ANOVA was violated and the results should be interpreted cautiously.